ClassSync: Project Prototype

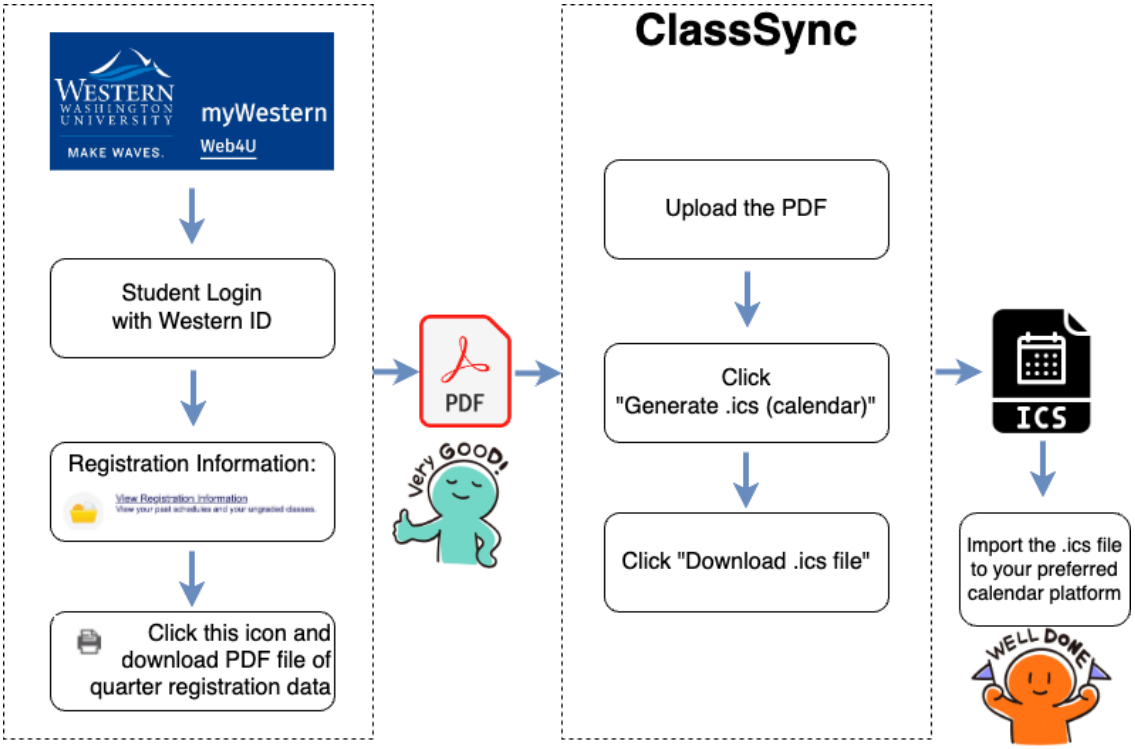
Team:

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About ClassSync:

ClassSync is a single-page web application that allows WWU students to manage their quarter-based school schedule effectively with an organized calendar. On ClassSync, students can upload a PDF file generated by the registration page on Web4U to generate a corresponding .ics (calendar) file.

Students can import this calendar file (.ics) to a calendar platform such as Outlook/Google Calendar/iCalendar to track their class schedule. Currently, students cannot access this quarterly calendar generation feature and ClassSync aims to make it accessible for everyone.



Need Finding (Student Interviews):

For need-finding, we designed and developed a set of eight interview questions for students. The interviews were conducted in person and online, keeping the students' availability in mind. We had five student interviewees (WWU undergraduate: 4, non-WWU graduate: 1).

Summary of Interview Findings:

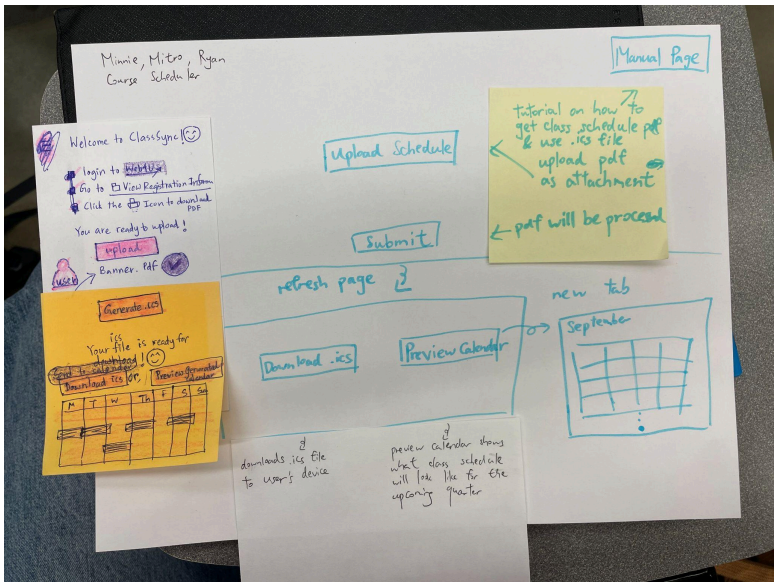
In our interviews, participants stated that they use multiple apps to manage their schedules: LMS calendars for assignment due dates, Web4U for class schedules, and other calendars (e.g. Outlook Calendar, Apple Calendar) for personal events. However, they prefer to have their class schedule and personal schedule combined into one calendar to have a better overview of their day.

Current systems do not allow these different schedules to be integrated, which they find inconvenient. Therefore, our system aims to address this problem by importing class schedules from Web4U into the Outlook calendar. Below are the detailed interview findings and features in our prototype that accommodate student's needs:

Key Findings	Details	Feature
Web4U: Inability for exportation	Web4U does not allow students to download their class timetable or add personal events to the schedule. Students can only take screenshots of it.	Class schedule pdf to .ics converter
Web4U: Inexperience in functionalities	Some participants stated that Web4U is cumbersome to use. Most are unaware that we can download a PDF of a worded version of our class schedule from Web4U.	Instructions to download the class schedule file
Web4U: Unclear representation of the class schedule	Web4U's class timetable only shows class times for the week. Participants reflected they would like to see what their class schedule will look like throughout the quarter in calendar view.	Calendar preview
LMS calendar usage	Students rely on the calendar feature in learning management systems (LMS), like Canvas or Moodle, for course scheduling and assignment management. They use these tools because all course information can be found in one place. However, they wish for more customization options and consistency in how schedules are organized.	Canvas calendar sync to Outlook (omitted)

Due to scope and time limitations, we were unable to include Canvas integration, but we addressed the user need for a combined schedule by designing features to import the Web4U class timetable into the Outlook calendar.

ClassSync (Paper Prototype):



As a preliminary design, the whole team designed ClassSync individually (3 designs) during an in-class group exercise.

After that, the team discussed their designs and finalized the sketch with a single design with specifications where all the features and usability points were present.

A paper-based low-fidelity prototype of ClassSync was submitted after the team discussion.

Brainstorming:

- (Leftmost vertical phone-based UI with an orange sticky note beneath)
  Phone screen with the 1-button design where the next buttons will appear (in the orange sticky note) after the PDF upload.
- (A4 white page with a yellow sticky note on the right)
  A two-page design with two buttons “Download .ics” and “Preview Calendar” will appear; “Preview Calendar” will open a new tab with the generated calendar preview upon clicking.

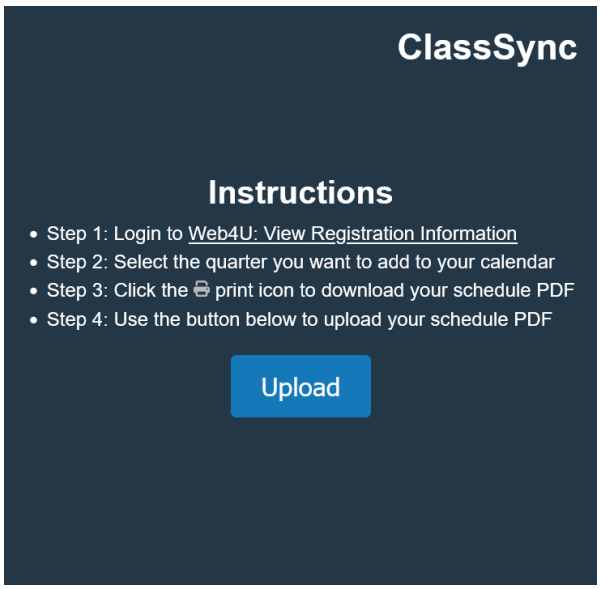
Team Discussion:

- During initial designs, all three team members limited the UI to 1-2 menus/options only. The first idea was to add a manual page separately to guide the students on how to get the PDF file from Web4U, but it was later decided to add 3-step points right above the “Upload” button.
- Instead of having multiple pages, it was decided that it’s optimal to keep all the buttons on the same page one by one in an ordered way of operation. One button is clicked and then a new button appears right after to proceed to the next action.
- The team decided to design a web app instead of a browser extension keeping two important points in mind,
  - **a)** Letting the students use ClassSync easily both on the Web and Mobile;
  - **b)** Understanding the practicality of technical scope within a few weeks;

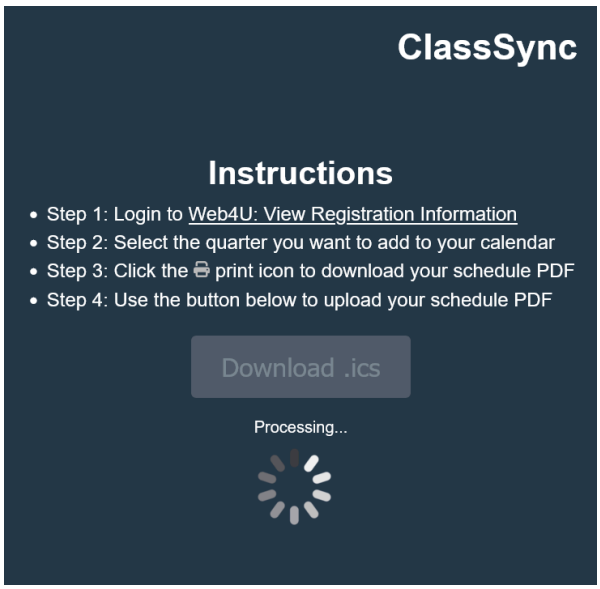
Scopes of ClassSync (Web UI Prototype):

After a thorough discussion on the finalized paper prototype, the team designed the UI using HTML and CSS while collaborating on a GitHub repository.

1. **[Feature Detail]** ClassSync is a single-page web application, which means all the functionalities will be handled in a single UI with dynamic appearance layouts. Students do not need to switch between several menus or UIs. The key is to make the system easy-to-use for students and reduce cognitive labor.



1. Opening UI of "ClassSync"



2. UI after uploading while file is being processed



3. UI after Successful PDF upload

Details: After opening ClassSync, students will see a simple page with one "Upload" button along with 4 steps and instructions on how to get the PDF file. Upon successful file upload, a new button "Download .ics" will appear automatically for the users to click and download their generated .ics file. This includes a slight change from the original paper prototype, where a "Generate .ics" button was included, but removed after being deemed unnecessary.

2. **[Feature Detail]** On ClassSync, students can upload a PDF of their class schedule from Web4U and generate a .ics file, allowing them to import it into their Outlook calendar. This allows the students to manage their class schedules for the entire quarter with a single import .ics file to their desired calendar platform.

Currently, there is no way to see the classes in the Canvas Calendar or the WWU Outlook calendar.

3. **[Feature Detail]** 4-step instructions on the same page with the "Upload" button to assist students on how to get the PDF file from Web4U. Later, there will be another simple step-by-step instruction on how to import the .ics file to the Outlook calendar after the .ics file is generated.

This will help students with easy-to-understand instructions on how to use ClassSync.

4. **[Feature Detail]** ClassSync also features a calendar preview of the generated .ics file for students to view their class schedules easily. This gives a good visualization of what their upcoming quarter's schedule will look like, which is better than Web4U's class schedule, which only shows what a week of class looks like.

5. **[Limitation & Future Work]** Currently, Canvas calendars hold the frequently added and updated submission deadlines set by the Professor. However, these dynamic deadline events do not get automatically imported to the WWU Outlook calendar. The calendar .ics file from Canvas can be later imported just like our generated .ics calendar.

We initially planned to get the .ics file from Canvas using the Canvas LMS REST API and merge the Canvas .ics with our generated quarter schedule .ics. As Canvas .ics is dynamically updated each time the Professor adds a new deadline, it can be checked in intervals for the updated schedule via API. However, we did not include this feature in ClassSync, because:

a. Ensuring proper authentication on Canvas for the WWU student accounts using the REST API can be challenging in a project with a short duration.

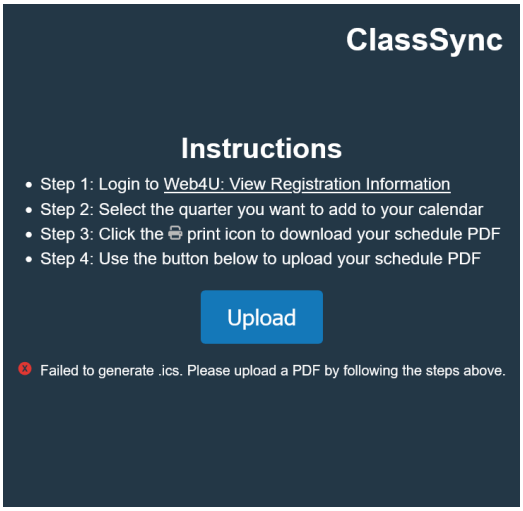
b. The current ClassSync tech stack (data parsing from PDF files and then generating .ics file) is also pretty interesting and complicated at the same time.

c. The team wanted to keep the project scopes practical within the designated time frame. We also believe the dynamic calendar events for submission deadlines would be very helpful. However, due to no proper institutional authentication between WWU Canvas and WWU Outlook, it can be a potential future work for us to improve the system.

ClassSync & Universal Design Principles:

The team collectively tried to incorporate the universal design principles into the ClassSync prototype:

Equitable: Keeping our target users WWU students in mind, we have attempted good web design practices such as easy-to-understand and legible text-based terms, high contrast and single-page navigation.
Flexible: ClassSync is flexibly displayed on the screens based on the different types of devices such as desktop or mobile.
Simple: ClassSync has a one-button-with-instructions approach to ease the cognitive burden of users for the next steps or any navigation.
Perceptible: All the texts are displayed in a high contrast and clean UI format along with precise instructions regardless of the user's experience.
Error Tolerance: ClassSync deals with PDF file upload. Hence, the potential error instances can be, users uploading the wrong file type, and users uploading a PDF that causes an issue during data parsing in the backend. There will be a unique and descriptive error for each of these instances to let users know what went wrong without breaking the site.



Error in data parsing



Wrong file type

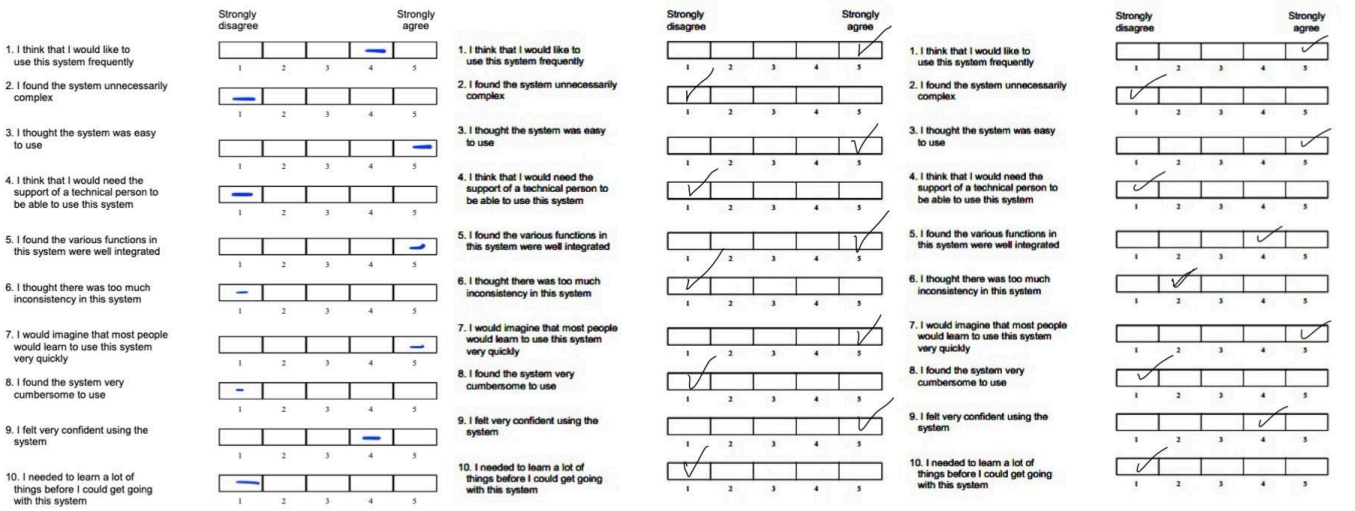
Easy to Use: To make it as clear as possible for users we have added a list of direct instructions to follow to find the PDF they can upload to ClassSync. The upload and download processes are controlled by a single button with no page refresh or page redirection.

Size and Space: It is a simple site where the content is centred on the screen of the device, allowing it to be used on desktop and mobile devices in the same capacity. All the content is designed to fit the different screen sizes.

System Usability Scale (SUS) Evaluation for ClassSync:

To understand user perception of the usability of our system, we asked 3 users from our target group (WWU students) to perform the SUS evaluation on our prototype.

User "X"	User "Y"	User "Z"



Based on research on SUS evaluation, we calculated the user response into a score using a SUS Calculator* and assigned it to grades A- F. A score above 80.3 is an A and anything below 51 is an F. Grade A indicates that the user thinks the system is usable and learnable and F is an unusable system.

SUS Scores: ClassSync			
Student User	"X"	"Y"	"Z"
Score	95	100	92.5
Grade	A	A	A

*Reference: SUS Calculator - UIUX Trend

Looking at the SUS grades, it appears that users are most likely to use ClassSync and also recommend it to a friend.